Highlights

Fine–Coarse Fingerprints and Temporal Complexity of Particulate Matter in the U.S.: A Land-Use and Seasonal Analysis with the Fine Dominance Index and Ordinal-Pattern Entropy

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- FDI and IFS fingerprint fine versus coarse PM dominance at 240 U.S. monitors
- Winter fine-dominance hotspots recur in suburban and urban built environments
- Daily PM series are near-noise yet ordered; entropy rises from $PM_{2.5}$ to PM_c
- Land use shapes PM_{10} but not $PM_{2.5}$ dynamics, which vary mainly with geography
- Fine-dominated sites show more structured $PM_{2.5}$ and less bursty PM_{10} dynamics

Fine–Coarse Fingerprints and Temporal Complexity of Particulate Matter in the U.S.: A Land-Use and Seasonal Analysis with the Fine Dominance Index and Ordinal-Pattern Entropy

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ABSTRACT

Particulate matter mixtures vary across space, seasons, and land uses, and a single $PM_{2.5}/PM_{10}$ ratio captures neither the size partition nor the temporal dynamics that distinguish source regimes. Using EPA AirData daily files for 2023–2025, we combine two layers. At 240 monitors with concurrent $PM_{2.5}$ and PM_{10} records, we compute a bounded Fine Dominance Index (FDI), summarize land use × location setting × season cells with a standardized composite index (IFS) and station-block bootstrap intervals, and test land-use contrasts on station–season–year units with Kruskal–Wallis tests, Holm-adjusted Mann–Whitney tests, and Cliff's δ. At 1066 monitors, we characterize the daily dynamics of $PM_{2.5}$, PM_{10}, and coarse PM (PM_c) with permutation entropy, statistical complexity, sample and SVD entropies, fractal dimensions, and surrogate tests.

Fine dominance is mainly seasonal: station FDI peaks in winter at 83% of monitors (Kendall's $W = 0.42$), the robust hotspots are winter cells in suburban and urban built environments, and coarse dominance prevails at rural industrial and desert sites. Once pseudo-replication is avoided, land-use contrasts within settings are mostly small. Daily PM series are high-entropy, correlated-noise processes that remain significantly more ordered than shuffled surrogates; ordinal entropy increases from $PM_{2.5}$ to PM_{10} to PM_c, and summer $PM_{2.5}$ dynamics are smoother but more heavy-tailed. Land use explains a moderate share of PM_{10} complexity ($\varepsilon^2 = 0.13$–0.15) but little of $PM_{2.5}$, whose dynamics vary mainly with geography. Fine-dominated stations show more structured $PM_{2.5}$ and less bursty PM_{10} dynamics ($|\rho_s| = 0.31$–0.35). Relying only on routinely reported mass concentrations, the framework is scalable and reproducible, and it indicates where chemical speciation and meteorological attribution would be most informative.

1. Introduction

Air pollution remains a critical environmental and public health concern in the United States and worldwide. Its complexity arises from the interplay among emission sources, land-use patterns, and meteorology, which together shape the spatial and temporal variability observed at monitoring networks. The U.S. Environmental Protection Agency (EPA) operates reference stations that track criteria pollutants, including particulate matter with aerodynamic diameters below 10 µm (PM_{10}) and 2.5 µm ($PM_{2.5}$), across urban cores, suburban belts, and rural areas, and across land-use contexts that range from commercial, residential, and industrial districts to agricultural land, deserts, military reservations, and forests.

Stratifying observations by location and land use sharpens inference on dominant sources and on exposure disparities (Kilpatrick et al., 2024; Hart et al., 2009; Sooktawee et al., 2020). Agricultural regions contribute substantial ammonia that promotes secondary $PM_{2.5}$ formation, with rising NH_3 emissions linked to confined animal feeding operations (Burns et al., 2023; Ogwu et al., 2024). Urban corridors are persistently influenced by traffic-related emissions (Alvarado-Molina et al., 2023; Nasr and Eissa, 2025), while industrial and residential zones display distinct particulate and gaseous profiles (Bikis, 2023; Roy et al., 2024). Seasonal cycles further modulate concentrations through boundary-layer dynamics, residential heating, photochemistry, wildfires, and dust intrusions

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that can episodically elevate particulate levels across large regions (Achilleos et al., 2020; Velásquez-García et al., 2024; Rawat et al., 2024; Butler and Hanigan, 2024; Yu et al., 2021). In parallel, the growing availability of regulatory-grade monitors and complementary low-cost sensors has improved spatial coverage and temporal resolution, albeit with calibration and comparability challenges (Chen et al., 2022; Brilli et al., 2021).

The fine–coarse partition is commonly characterized through the $PM_{2.5}/PM_{10}$ ratio, which is typically higher in combustion- and traffic-influenced environments and lower where resuspension or natural dust dominates (Fan et al., 2021; Colangeli et al., 2022). While informative, single-ratio summaries and fixed thresholds can mask heterogeneity across land uses and along urban–suburban–rural gradients, where long-term patterns and disparities also emerge (Kilpatrick et al., 2024; Liu and Marshall, 2023; Jbaily et al., 2022). More fundamentally, ratio-based summaries are static: two monitors with identical seasonal ratios can differ sharply in how concentrations evolve from one day to the next, that is, in their persistence, their predictability, and the intermittency of their excursions. These dynamical properties carry information about the processes that generate the signal, because regionally mixed secondary aerosol, local resuspension, and episodic dust or smoke events leave different temporal fingerprints.

Information-theoretic and fractal descriptors offer a compact way to quantify such fingerprints. Permutation entropy summarizes the diversity of ordinal patterns in a time series (Bandt and Pompe, 2002), and its combination with the Jensen–Shannon statistical complexity in the complexity–entropy plane separates stochastic from structured dynamics (Lamberti et al., 2004; Rosso et al., 2007; Martin et al., 2006); sample entropy measures amplitude-based irregularity (Richman and Moorman, 2000), and the Higuchi and correlation dimensions characterize the roughness and geometric complexity of a trajectory (Higuchi, 1988; Grassberger and Procaccia, 1983). Applications to air quality have shown that the disorder of pollutant time series differs between traffic-related and rural stations (Amato et al., 2020) and that long PM_{10} records combine stochastic and self-organized components (Nikolopoulos et al., 2021). To our knowledge, however, these descriptors have not been combined with fine–coarse partition indices across a national regulatory network stratified by land use, location setting, and season.

In a companion study, we introduced a nonlinear, robust station–season index (S_{NLR}) that integrates the median fine fraction, the typical $PM_{2.5}$ level, and coarse-tail burstiness for 2022–2024, and found that seasonality dominates over land-use contrasts (Sierra-Porta et al., 2026). The present work is complementary in three respects. First, we work at the scale of individual daily samples and land use–location–season cells through a Fine Dominance Index (FDI), designed to increase with the predominance of fine particles, and a cell-level composite score (IFS) that standardizes multiple cell summaries into a single fingerprint. Second, we add a temporal-complexity layer that describes how $PM_{2.5}$, PM_{10}, and coarse PM (PM_c) concentrations evolve, rather than how large they are. Third, we update the analysis to the most recent complete three-year window (2023–2025), implement all inferential contrasts on station-level units to avoid pseudo-replication (Hurlbert, 1984), and test the robustness of every index to weighting, screening, and quality-control choices. We address three questions: (i) where and when fine particles dominate the PM_{10} mass; (ii) whether daily PM dynamics differ among size fractions, seasons, land uses, and regions; and (iii) whether the size partition and the temporal dynamics are related at the station level.

The scope of this study is deliberately limited to mass concentrations that are routinely reported across the network. Indices derived from PM_{10} and $PM_{2.5}$ alone can be applied at scale without additional instrumentation, but they remain indirect: ratios are sensitive to measurement artifacts and shared-denominator effects, and neither ratios nor entropies separate individual sources. We therefore combine several complementary metrics, stratify all analyses by land use and location setting, quantify uncertainty with resampling, and report effect sizes alongside multiplicity-adjusted tests. The resulting evidence is intended for screening and prioritization, and for identifying where chemically resolved measurements and meteorological attribution would add the most information.

The remainder of this article is organized as follows. Section 2 describes the data, the quality-control steps, and the construction of the derived indices, complexity metrics, and statistical tests. Section 3 presents the distributional and cell-wise fine–coarse fingerprints, the formal contrasts and robustness checks, the temporal-complexity analysis, and the link between both layers, followed by a discussion of implications and limitations. Section 4 summarizes the conclusions and directions for future work.

2. Data and methods

2.1. Data

We retrieved the pre-generated daily summary archives of the EPA AirData portal (`https://aqs.epa.gov/aqsweb/airdata/download_files.html`) for 2023, 2024, and 2025 and for four parameters: $PM_{2.5}$ FRM/FEM mass (parameter

Table 1
Open sites in the AQS inventory for the 50 U.S. states and the District of Columbia (n = 4652), by land use and by location setting, as reported in the site metadata.

Land use	Count
Residential	1617
Commercial	1077
Industrial	701
Agricultural	587
Forest	436
Desert	104
Mobile	67
Unknown	47
Military reservation	13
Blighted areas	3

Location setting	Count
Rural	1830
Suburban	1589
Urban and center city	1204
Unknown	29

code 88101; `daily_88101_YYYY.zip`), $PM_{2.5}$ non-FRM/FEM mass (88502; `daily_88502_YYYY.zip`), PM_{10} mass (81102; `daily_81102_YYYY.zip`), and coarse particulate matter PM_c ($PM_{10-2.5}$; 86101; `daily_86101_YYYY.zip`), together with the site metadata file (`aqs_sites.zip`), which provides the land-use class and location setting of every monitoring site. Each daily record reports the *Arithmetic Mean* of the valid sub-daily measurements of one instrument, identified by the AQS site code (State Code, County Code, Site Number) and the Parameter Occurrence Code (POC).

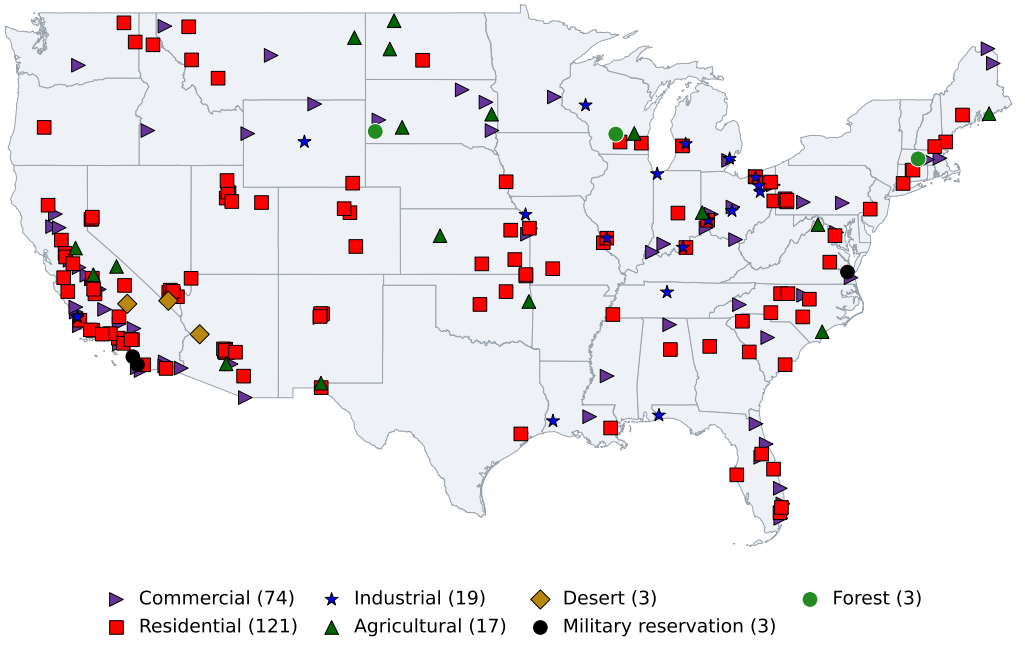
Because the daily files contain several records per site and day, we applied the following deterministic reduction before any analysis. Records flagged as *Excluded*, which repeat a monitor-day after removing data affected by exceptional events, were discarded so that each day retains the measured concentration, and daily summaries built from less than 75% of the scheduled sub-daily observations were also removed. Identical values repeated once per pollutant standard were collapsed, and, for each instrument and day, a single sample duration was retained with the priority 24-h block average > 24-h sample > 1-h aggregate. Finally, because averaging collocated instruments with different sampling schedules (for example, 1-in-3-day filter samplers and continuous monitors) injects artificial day-to-day jumps into a time series, each site retained the single most complete instrument record (POC) for each parameter. This reduction removes up to ten records per site and day in the $PM_{2.5}$ FRM/FEM files and yields one daily value per site, parameter, and day. The analysis is restricted to the 50 states and the District of Columbia, to the land-use classes commercial, residential, industrial, agricultural, desert, military reservation, and forest, and to the location settings rural, suburban, and urban and center city; mobile, blighted, and unknown classes were excluded because of their low counts or ambiguous interpretation. $PM_{2.5}$ FRM/FEM and PM_{10} form the primary regulatory series; $PM_{2.5}$ non-FRM/FEM and PM_c are analyzed in the complexity layer and serve as consistency checks.

Network composition and completeness

Table 1 summarizes the open U.S. sites in the AQS inventory by land use and by location setting, as reported in the site metadata. Because many monitors report intermittently, we applied a completeness screen at the station–parameter level: a station was retained for a given parameter when it reported at least 822 valid daily values over the 1096 days of 2023–2025 (75% coverage; 2024 is a leap year). For the fine–coarse layer, the same criterion was applied to the number of days with concurrent $PM_{2.5}$ FRM/FEM and PM_{10} values, because every ratio requires same-day availability of both measurements.

After screening, the fine–coarse network comprises **240** monitors with 247,270 station-days of concurrent $PM_{2.5}$ and PM_{10} data, distributed across the seven land-use classes and the urban–suburban–rural gradient (Table 2). The temporal-complexity network is larger because it requires only one parameter at a time: 751 monitors for $PM_{2.5}$ FRM/FEM, 171 for $PM_{2.5}$ non-FRM/FEM, 420 for PM_{10}, and 91 for PM_c, that is, 1066 distinct monitors. Desert, forest, and military reservation sites are sparse in the fine–coarse network (three stations each), whereas the PM_{10} complexity network includes 25 desert monitors, which allows the coarse-dominated regime to be characterized dynamically with adequate replication. Figure 1 shows the geographic distribution of both networks.

(a) Fine Dominance Index network, n = 240 stations (4 outside the map extent)



(b) Temporal-complexity network, n = 1066 stations (21 outside the map extent)

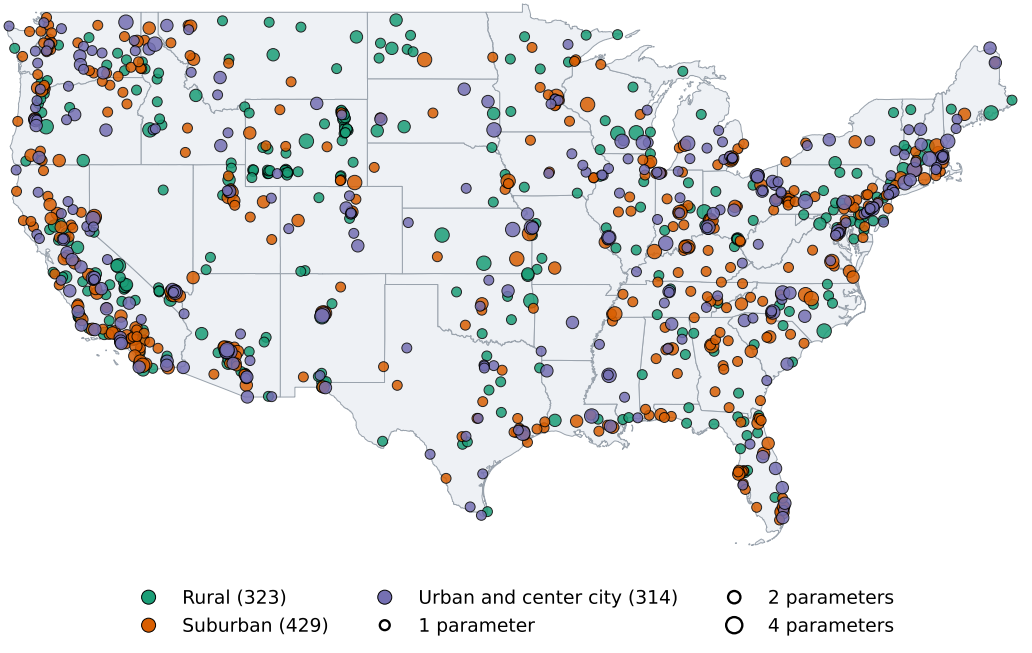


Figure 1: Retained monitoring stations in the contiguous United States. (a) Fine–coarse (FDI) network ($n = 240$), colored by land use; four stations in Alaska and Hawaii lie outside the map extent. (b) Temporal-complexity network ($n = 1066$; 21 stations outside the map extent), colored by location setting; marker size indicates the number of parameters ($PM_{2.5}$ FRM/FEM, $PM_{2.5}$ non-FRM/FEM, PM_{10}, PM_c) that passed the completeness screen at each site.

Table 2
Retained stations (≥822 valid days in 2023–2025) by land use. The fine–coarse network requires concurrent $PM_{2.5}$ FRM/FEM and PM_{10} records and is broken down by location setting; the complexity networks are screened separately for each parameter.

	Fine–coarse (FDI) network				Temporal-complexity networks			
Land use	**Rural**	**Suburban**	**Urban and center city**	**Total**	**$PM_{2.5}$ FRM/FEM**	**$PM_{2.5}$ non-FRM/FEM**	**PM_{10}**	**PM_c**
Commercial	6	22	46	74	226	45	105	31
Residential	9	70	42	121	351	90	170	44
Industrial	4	7	8	19	61	9	84	4
Agricultural	17	0	0	17	75	21	29	7
Desert	2	0	1	3	7	1	25	0
Military reservation	0	3	0	3	4	0	3	0
Forest	3	0	0	3	27	5	4	5
Total	**41**	**102**	**97**	**240**	**751**	**171**	**420**	**91**

2.2. Methods

2.2.1. Seasons and units of analysis

Climatological seasons were defined by calendar months: winter (December–February), spring (March–May), summer (June–August), and autumn (September–November). For pooled seasonal summaries, each daily record inherits the season of its date. For analyses that require contiguous periods or year-specific units, we used season–years in which December is assigned to the following winter; thus, winter 2023 spans January–February 2023, winter 2024 spans December 2023–February 2024, and December 2025, which belongs to winter 2026, is not used. Daily values from the same monitor are serially correlated, so treating them as independent replicates inflates the significance of between-group tests (Hurlbert, 1984; Sierra-Porta et al., 2026). All hypothesis tests are therefore carried out on station-level units (station–season–year means for the fine–coarse layer and station medians for the complexity layer), whereas daily values are used only for descriptive distributions.

2.2.2. Fine Dominance Index

For each station-day with positive $PM_{2.5}$ and PM_{10}, we computed three ratios that capture complementary aspects of the fine–coarse partition (Fan et al., 2021),

$$r_{12} = \frac{PM_{2.5}}{PM_{10}}, \qquad r_{21} = \frac{PM_{10}}{PM_{2.5}}, \qquad r_0 = \frac{PM_{10} - PM_{2.5}}{PM_{10} + PM_{2.5}}, \tag{1}$$

together with the per-sample Fine Dominance Index (FDI),

$$FDI = \frac{PM_{2.5}}{PM_{10}}(1 - r_0) = \frac{2\,r_{12}^2}{1 + r_{12}}, \tag{2}$$

which increases with fine-particle dominance. Because $PM_{2.5}$ is a subset of PM_{10}, days with $PM_{2.5} > PM_{10}$ are physically inconsistent and reflect instrument differences or measurement uncertainty; they represent 1.6% of the station-days with positive values and were excluded. With $0 < r_{12} \leq 1$, the FDI is bounded in $(0, 1]$ and r_0 in $[0, 1)$, so that neither index is affected by the near-zero denominators that make unbounded ratios heavy-tailed. The final fine–coarse dataset contains 241,664 station-days. Equation (2) shows that the FDI is a monotone and convex transformation of r_{12}: it preserves the ranking of days while expanding differences among fine-dominated days, which is why rank-based tests on daily FDI, r_{12}, r_{21}, and r_0 are equivalent and only the FDI is reported.

2.2.3. Cell-wise fingerprints and the composite index IFS

For interpretability across contexts, observations were summarized in cells defined by the cross of land use, location setting, and season (56 populated cells). Within each cell we computed the mean of the FDI ($\overline{FDI}$), the mean of r_0 ($\overline{r_0}$), and the skewness and excess kurtosis of r_0 (Fisher definitions). To provide a single fingerprint per cell, we built the

composite index

$$\mathrm{IFS} = w_{\mathrm{FDI}}\, z\big(\overline{\mathrm{FDI}}\big) + w_{r_0}\, z\big(\overline{r_0}\big) + w_{\mathrm{skew}}\, z\big(\mathrm{skew}(r_0)\big) + w_{\mathrm{kurt}}\, z\big(\mathrm{kurt}(r_0)\big), \tag{3}$$

where $z(\cdot)$ denotes z-scores computed across all populated cells, and $w_{\mathrm{FDI}} = 1.5$, $w_{r_0} = -1.0$, $w_{\mathrm{skew}} = -0.5$, and $w_{\mathrm{kurt}} = -0.5$. The weighting emphasizes fine dominance via $\overline{\mathrm{FDI}}$, penalizes coarse dominance via $\overline{r_0}$, and assigns a modest penalty to right-skewed, heavy-tailed r_0 distributions, which signal episodic coarse bursts. Because the components are standardized across cells, the IFS is on a z-scale: values near zero reflect network-average behavior, positive values indicate cells that are more fine-dominated than the network average with relatively muted coarse bursts, and negative values indicate comparatively coarse-dominated or bursty regimes.

Uncertainty in $\overline{\mathrm{FDI}}$ was quantified with a station-block bootstrap ($B = 2000$) that resamples whole stations within each cell and therefore preserves the serial dependence of daily data (Efron and Tibshirani, 1994). A cell was flagged as thin when it contained fewer than three stations, fewer than 300 station-days, or a 95% bootstrap interval wider than 0.15 FDI units; thin cells are hatched in the heatmaps and excluded from the ranked tables. Of the 56 populated cells, 38 are not thin.

The robustness of the IFS was evaluated against (i) a grid of $5^4 = 625$ weight combinations ($w_{\mathrm{FDI}} \in [1.0, 2.0]$, $w_{r_0} \in [-1.5, -0.5]$, $w_{\mathrm{skew}}, w_{\mathrm{kurt}} \in [-1.0, 0]$); (ii) named alternatives, including equal weights, location terms only, shape terms with a positive sign, and the mean FDI alone; (iii) the first principal component of the four standardized features; (iv) a stricter completeness threshold of 90% (987 days); (v) the retention of days with $\mathrm{PM}_{2.5} > \mathrm{PM}_{10}$ after capping r_{12} at 1; (vi) alternative thin-cell criteria; and (vii) a separate recomputation for each season–year. Agreement with the nominal ranking of non-thin cells is reported as Kendall's τ and as the overlap of the top-10 and bottom-10 league tables.

2.2.4. Formal contrasts

Land uses were compared within each location setting and season on station–season–year means of the FDI, computed from units with at least 45 ratio days. Kruskal–Wallis omnibus tests (Kruskal and Wallis, 1952; Lachenbruch and Clements, 1991; MacFarland and Yates, 2016) included land uses with at least three stations, and the resulting family of omnibus tests (12 strata × four variables: FDI, r_0, $\mathrm{PM}_{2.5}$, and PM_{10}) was adjusted with the Holm procedure (Holm, 1979). The omnibus effect size is reported as $\varepsilon^2 = H/(n-1)$ (Kelley, 1935). Pairwise post hoc contrasts used two-sided Mann–Whitney tests (Mann and Whitney, 1947; Nachar, 2008) with Holm correction within each stratum, and magnitudes were summarized with Cliff's δ (Cliff, 1993; Meissel and Yao, 2024), interpreting $|\delta| < 0.147$ as negligible, $0.147 \le |\delta| < 0.33$ as small, $0.33 \le |\delta| < 0.474$ as medium, and $|\delta| \ge 0.474$ as large (Romano et al., 2006). The seasonal contrast within stations was evaluated with the Friedman test on station-level seasonal means of the FDI (Friedman, 1937), with Kendall's W as effect size (Kendall and Babington Smith, 1939).

2.2.5. Temporal complexity of daily PM series

For every station and parameter of the complexity network, the daily series was partitioned into season–year windows (12 windows per station). A window was analyzed when at least 75% of its calendar days had data; gaps of up to three days were linearly interpolated, and longer gaps were left unfilled. Each window (on average 90 days) was standardized, and ties, which are frequent because PM_{10} is reported with a resolution of $1\,\mu\mathrm{g\,m^{-3}}$, were broken at random with a negligible jitter, since equal values otherwise bias ordinal-pattern statistics (Zunino et al., 2017). A total of 16,411 windows were analyzed (8664 for $\mathrm{PM}_{2.5}$ FRM/FEM, 1936 for $\mathrm{PM}_{2.5}$ non-FRM/FEM, 4782 for PM_{10}, and 1029 for PM_c).

The metrics, summarized in Table 3, follow established definitions. For ordinal patterns of embedding dimension D and delay τ with probability distribution $P = \{p_i\}_{i=1}^{D!}$, the normalized permutation entropy is $H[P] = -\sum_i p_i \ln p_i / \ln D!$ (Bandt and Pompe, 2002), and the Jensen–Shannon statistical complexity is $C_{JS}[P] = Q_J[P, P_e]\, H[P]$, where Q_J is the normalized Jensen–Shannon divergence between P and the uniform distribution P_e (Lamberti et al., 2004; Rosso et al., 2007). Sample entropy is $-\ln(A/B)$, where B and A count pairs of templates of length m and $m+1$ that match within a tolerance r (Richman and Moorman, 2000). SVD entropy is the normalized Shannon entropy of the singular values of the embedding matrix (Roberts et al., 1999); the Higuchi fractal dimension is the slope of the log–log relation between curve length and scale (Higuchi, 1988); and the correlation dimension D_2 is the scaling exponent of the Grassberger–Procaccia correlation sum (Grassberger and Procaccia, 1983), estimated by RANSAC regression with a fixed random seed for reproducibility. Skewness and excess kurtosis of the concentrations

Table 3
Temporal-complexity metrics computed for each station–parameter–season–year window. The full-period ordinal analysis uses $D = 4$ on the complete 2023–2025 record of each station.

Metric	Settings	Range	High values indicate
Permutation entropy H	Ordinal patterns, $D = 3$ (windows) or $D = 4$ (full period), $\tau = 1$ day	$[0, 1]$	Ordinal sequences closer to uncorrelated noise
Statistical complexity C_{JS}	Same ordinal distribution as H	$[0, C_{\max}(H)]$	Structured, correlated dynamics that are neither fully ordered nor random
Sample entropy	$m = 2$, $r = 0.2\times$SD	≥ 0	Amplitude irregularity (few repeated templates)
SVD entropy	Embedding order 3, $\tau = 1$ day	$[0, 1]$	More independent components in the embedded trajectory
Higuchi fractal dimension	$k_{\max} = 10$	$[1, 2]$	Rougher trajectories (1.5 for Brownian motion, 2 for white noise)
Correlation dimension D_2	Embedding dimension 2	$[0, 2]$	Space-filling of the reconstructed state space (bounded by the embedding)
Skewness, excess kurtosis	Fisher definitions	$\mathbb{R}$	Asymmetric, heavy-tailed (bursty) concentration distributions

describe the asymmetry and tail weight of each window. Permutation entropy, SVD entropy, and the Higuchi dimension were computed with `antropy` (0.2.2), sample entropy and the correlation dimension with `nolds` (0.6.3), and the statistical complexity with our own implementation of the Jensen–Shannon formulation.

The temporal structure of each window was tested against 199 shuffled surrogates, which preserve the distribution of values but destroy temporal order (Theiler et al., 1992); a window is considered ordered when its permutation entropy lies below the surrogate distribution at the 5% level. Because $D = 3$ (six patterns) is appropriate for windows of about 90 days but provides a coarse description of the complexity–entropy plane, we also computed H and C_{JS} with $D = 4$ (24 patterns) from the complete 2023–2025 record of each station, counting ordinal patterns only within contiguous data segments, and compared them with the bounds $C_{\min}(H)$ and $C_{\max}(H)$ of the plane (Martin et al., 2006) and with 99 shuffled surrogates per station.

Window-level metrics were summarized by their median per station (and per station and season). Differences among land uses and among location settings were tested with Kruskal–Wallis tests on station medians (groups with at least five stations; ε^2 as effect size), seasonal differences within stations with Friedman tests (Kendall's W), and differences between size fractions measured at the same site with Wilcoxon signed-rank tests (Wilcoxon, 1945), summarized by the matched-pairs rank-biserial correlation r_{rb} (Kerby, 2014). Holm corrections were applied within each family of tests. Redundancy among metrics was assessed with Spearman correlations across all windows, and the persistence of station rankings with Spearman correlations between years. Finally, the $PM_{2.5}$ FRM/FEM stations were grouped by Ward's hierarchical clustering (Ward, 1963) of the standardized medians of four non-redundant dynamical metrics (H, sample entropy, SVD entropy, and Higuchi dimension), with $k = 3$; C_{JS} was omitted because it is a deterministic function of H for $D = 3$, and D_2 because it is the least stable estimate. Cluster quality was assessed with the silhouette width and by reclustering 200 random subsamples of 80% of the stations, and agreement between clusters and land use or location setting was measured with the adjusted Rand index (Hubert and Arabie, 1985) and Cramér's V.

2.2.6. Linking the fine–coarse partition and temporal complexity

At the 240 stations of the fine–coarse network, the station mean FDI was correlated with the station-median complexity metrics of the collocated $PM_{2.5}$ FRM/FEM and PM_{10} series using Spearman's ρ_s, with Holm correction. Because both the size partition and the dynamics vary regionally, we also report partial Spearman correlations controlling for latitude and longitude.

2.2.7. Software and reproducibility

All analyses were performed in Python 3.13 with `pandas`, `numpy`, `scipy`, `scikit-learn`, `matplotlib`, `antropy`, and `nolds`. The pipeline downloads the public EPA AirData files, applies the reduction and screening rules described above, and reproduces every table and figure of this article with fixed random seeds.

3. Results and Discussion

3.1. Sample-level distributions of fine dominance

Figure 2 displays the daily FDI by land use, location setting, and season under a common y-axis. The dominant feature is seasonal. Station-level FDI peaks in winter at 83% of the monitors, with a network median of 0.35 in winter against 0.25 in spring, 0.23 in summer, and 0.24 in autumn (Friedman $\chi^2 = 301.6$, $p < 0.001$, Kendall's $W = 0.42$). The seasonal contrast is strongest in suburban ($W = 0.49$) and urban ($W = 0.43$) settings and weaker, though still significant, in rural settings ($W = 0.29$). This pattern is consistent with shallow wintertime boundary layers, residential heating, and the accumulation of secondary fine aerosol during stagnation, and with the larger contribution of resuspended and wind-blown coarse particles in the warm and dry seasons (Jandacka et al., 2024; Czernecki et al., 2021; Clements et al., 2016).

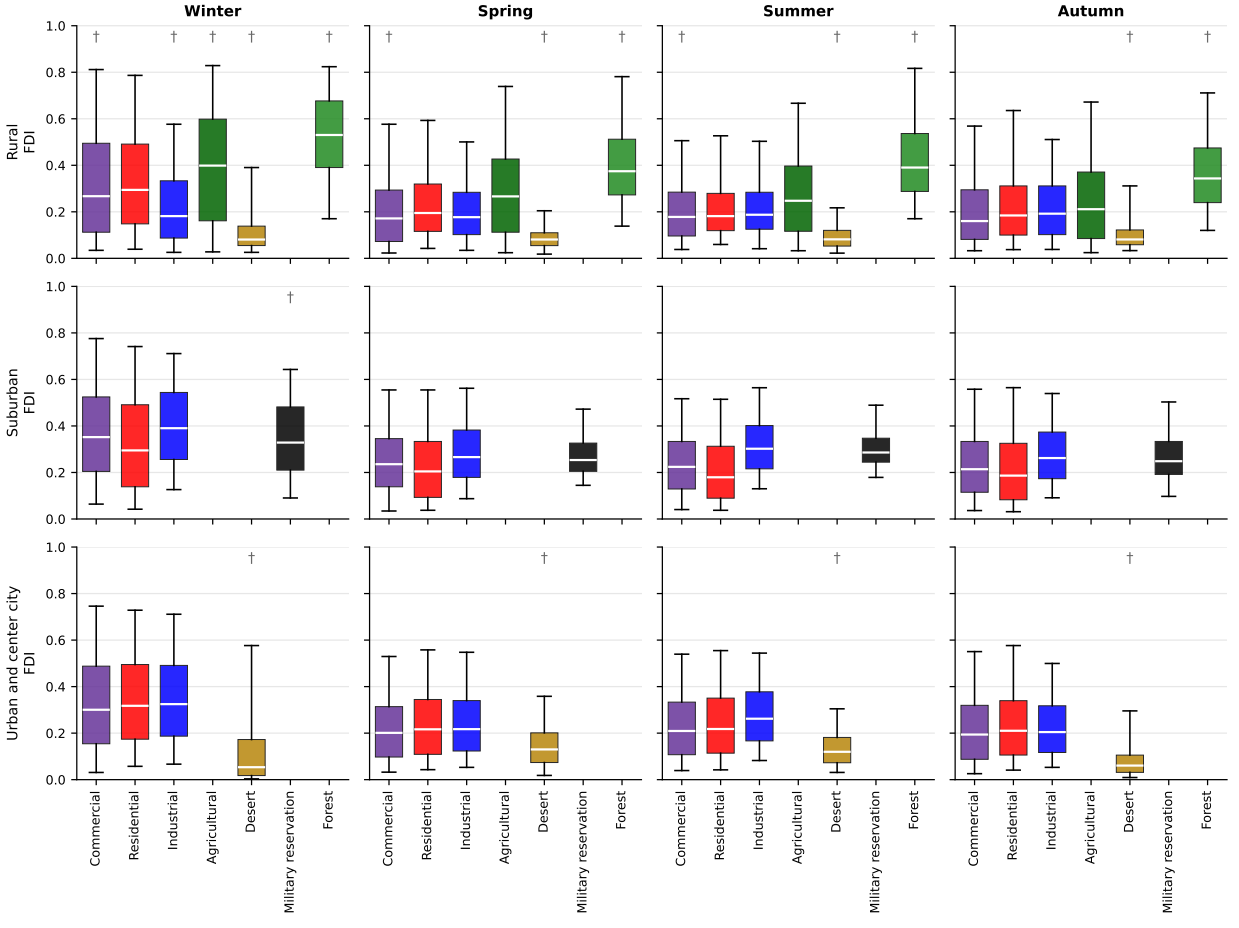


Figure 2: Daily Fine Dominance Index (FDI) by land use, with location setting in rows and season in columns. Boxes show medians and interquartile ranges, and whiskers span the 5th–95th percentiles; values outside the whiskers are not drawn. All panels share the y-axis. Daggers (†) mark thin cells (fewer than three stations, fewer than 300 station-days, or a 95% station-block bootstrap interval wider than 0.15). Larger FDI indicates stronger fine-particle dominance relative to coarse contributions.

Within each setting and season, the distributions of commercial, residential, and industrial monitors overlap broadly, with interquartile ranges that share most of their extent. The clearest separations involve the less common classes. Rural forest monitors are the most fine-dominated in every season (median FDI between 0.34 and 0.53), as expected for sites with little local resuspension that are periodically affected by smoke. Desert monitors are the most coarse-dominated, with medians below 0.15 in every setting and season, which reflects persistent soil dust. Agricultural monitors show a wide winter distribution and moderate values in the rest of the year, in line with the seasonal

contribution of secondary ammonium nitrate in agricultural regions (Burns et al., 2023; Bell et al., 2007). Higher FDI values point to combustion- and secondary-aerosol influence, whereas lower values indicate coarse dominance from resuspension or natural dust (Liu and Marshall, 2023; Xu et al., 2020); within a given setting, the contrasts among classes plausibly reflect activity patterns and surface characteristics (Wei and Wang, 2021; Clark and Dellachiesa, 2013). Where distributions overlap, the overlap suggests mixed-source environments or within-class heterogeneity, which motivates the cell-wise summaries and formal tests that follow.

3.2. Cell-wise seasonal fingerprints (IFS)

Figure 3 summarizes the composite IFS for each land use, location setting, and season under a common, zero-centered color scale. In suburban and urban settings, winter is the only season in which every built-environment class has a positive IFS (1.67–3.63), which identifies a coherent wintertime fine-dominance regime in commercial, residential, and industrial areas. Suburban industrial sites stand out with positive values in all seasons (1.11–3.63), and the three suburban military reservation sites are also fine-dominated outside winter. In rural settings, residential sites follow the winter pattern (1.94), agricultural sites are mildly positive in spring and summer (1.04 and 0.71), and industrial sites are the most coarse-dominated non-thin cells of the network in spring, summer, and autumn (-1.25 to -0.98). The extremes of the IFS scale correspond to thin cells: forest sites are strongly positive in all seasons (2.55–4.84), and desert sites are strongly negative in all seasons and settings (-5.65 to -3.14), in agreement with the boxplots but based on only one to three stations.

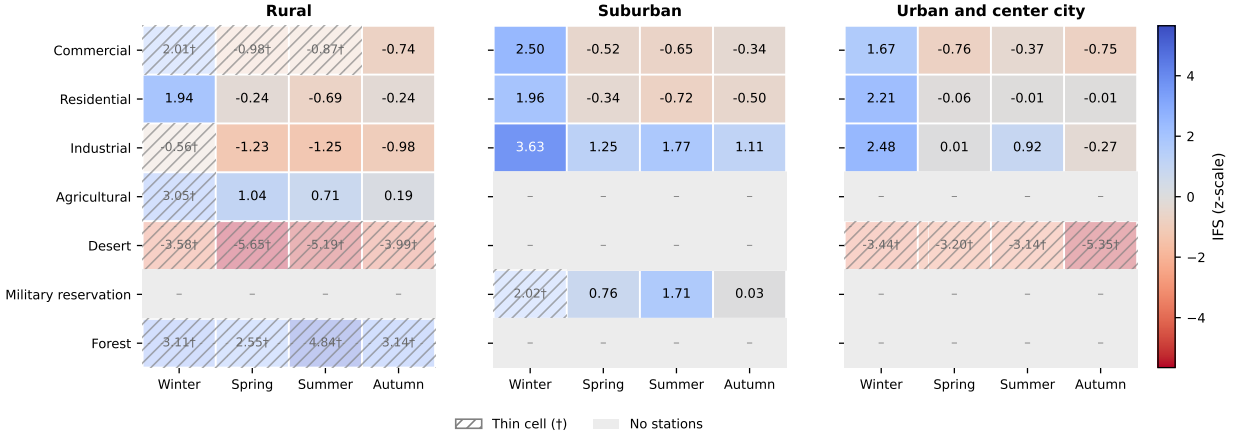


Figure 3: Cell-wise seasonal fingerprints given by the standardized composite index (IFS), by land use (rows) and season (columns), in three panels by location setting. The color scale is centered at zero and common to all panels. Hatched cells marked with a dagger (†) are thin and are excluded from the ranked tables; grey cells contain no stations. Positive values indicate stronger fine-particle dominance and more muted coarse bursts than the network-wide average across all cells.

Two points guide the interpretation. First, because the IFS is standardized across all cells and seasons, positive cells mark regimes that are more fine-dominated than the network-wide average, and negative cells mark comparatively coarse-dominated or bursty regimes; the values are therefore relative and should be read together with the physically interpretable FDI means. Second, outlying values often correspond to sparse classes or to seasons affected by episodic dust or smoke; when the number of stations is small or the bootstrap interval is wide, the corresponding cells are flagged rather than ranked.

3.3. Formal contrasts and ranked hotspots/coldspots

Table 4 ranks the non-thin cells by IFS. Seven of the ten hotspots are winter cells in built environments (suburban industrial, suburban commercial, urban industrial, urban residential, suburban residential, rural residential, and urban commercial), with mean FDI between 0.33 and 0.40, while the remaining three are suburban industrial and military reservation cells in summer and spring. The coldspots are dominated by rural industrial cells outside winter and by spring, summer, and autumn cells of commercial and residential sites, with mean FDI between 0.21 and 0.26 and mean r_0 close to 0.5. The skewness and kurtosis of r_0 are moderate in all ranked cells ($|\text{skew}| \leq 0.6$, $|\text{kurt}| \leq 0.8$), which

Table 4
Top 10 hotspots (highest IFS) and bottom 10 coldspots (lowest IFS) among the 38 non-thin cells. Columns report the cell mean of the FDI with its 95% station-block bootstrap interval, the number of contributing station-days (n) and stations (n_{st}), the composite index (IFS), and the mean, skewness, and excess kurtosis of r_0. Thin cells, including all desert and forest cells, are shown in Fig. 3.

Top 10 hotspots (highest IFS): most fine-dominated										
Land use	**Location setting**	**Season**	$\overline{\text{FDI}}$	**95% CI**	n	n_{st}	**IFS**	**Mean r_0**	**Skew r_0**	**Kurt r_0**
Industrial	Suburban	Winter	0.40	[0.36, 0.46]	1653	7	3.63	0.31	0.49	−0.31
Commercial	Suburban	Winter	0.38	[0.32, 0.43]	5262	22	2.50	0.34	0.59	0.06
Industrial	Urban and center city	Winter	0.35	[0.31, 0.40]	1788	8	2.48	0.36	0.36	−0.54
Residential	Urban and center city	Winter	0.35	[0.31, 0.38]	10402	42	2.21	0.37	0.39	−0.44
Residential	Suburban	Winter	0.33	[0.30, 0.36]	17343	70	1.96	0.39	0.31	−0.75
Residential	Rural	Winter	0.34	[0.28, 0.42]	2202	9	1.94	0.38	0.33	−0.47
Industrial	Suburban	Summer	0.32	[0.28, 0.35]	1867	7	1.77	0.36	0.22	−0.23
Military reservation	Suburban	Summer	0.30	[0.27, 0.37]	795	3	1.71	0.37	−0.30	0.50
Commercial	Urban and center city	Winter	0.33	[0.29, 0.37]	11229	46	1.67	0.39	0.48	−0.43
Industrial	Suburban	Spring	0.29	[0.25, 0.33]	1776	7	1.25	0.39	0.12	−0.53
Bottom 10 coldspots (lowest IFS): most coarse-dominated										
Land use	**Location setting**	**Season**	$\overline{\text{FDI}}$	**95% CI**	n	n_{st}	**IFS**	**Mean r_0**	**Skew r_0**	**Kurt r_0**
Industrial	Rural	Summer	0.22	[0.20, 0.25]	1009	4	−1.25	0.47	0.10	0.40
Industrial	Rural	Spring	0.21	[0.16, 0.27]	1002	4	−1.23	0.49	−0.04	−0.04
Industrial	Rural	Autumn	0.22	[0.17, 0.29]	933	4	−0.98	0.47	0.18	−0.29
Commercial	Urban and center city	Spring	0.23	[0.20, 0.25]	11535	46	−0.76	0.47	0.13	−0.47
Commercial	Urban and center city	Autumn	0.23	[0.20, 0.25]	11792	46	−0.75	0.48	0.09	−0.55
Commercial	Rural	Autumn	0.21	[0.14, 0.28]	1495	6	−0.74	0.50	−0.35	−0.50
Residential	Suburban	Summer	0.22	[0.19, 0.24]	18245	70	−0.72	0.48	−0.04	−0.69
Residential	Rural	Summer	0.22	[0.18, 0.27]	2291	9	−0.69	0.47	−0.35	0.28
Commercial	Suburban	Summer	0.24	[0.20, 0.29]	5580	22	−0.65	0.45	0.40	−0.21
Commercial	Suburban	Spring	0.26	[0.22, 0.29]	5278	22	−0.52	0.44	0.44	0.09

Table 5
Non-thin cells with positive IFS within each location setting (up to five per setting). Columns as in Table 4.

Land use	**Location setting**	**Season**	$\overline{\text{FDI}}$	**95% CI**	n	n_{st}	**IFS**	**Mean r_0**	**Skew r_0**	**Kurt r_0**
Rural										
Residential	Rural	Winter	0.34	[0.28, 0.42]	2202	9	1.94	0.38	0.33	−0.47
Agricultural	Rural	Spring	0.30	[0.23, 0.36]	4122	17	1.04	0.42	0.27	−0.70
Agricultural	Rural	Summer	0.28	[0.22, 0.34]	4397	17	0.71	0.43	0.20	−0.66
Agricultural	Rural	Autumn	0.26	[0.20, 0.31]	4304	17	0.19	0.46	0.04	−0.83
Suburban										
Industrial	Suburban	Winter	0.40	[0.36, 0.46]	1653	7	3.63	0.31	0.49	−0.31
Commercial	Suburban	Winter	0.38	[0.32, 0.43]	5262	22	2.50	0.34	0.59	0.06
Residential	Suburban	Winter	0.33	[0.30, 0.36]	17343	70	1.96	0.39	0.31	−0.75
Industrial	Suburban	Summer	0.32	[0.28, 0.35]	1867	7	1.77	0.36	0.22	−0.23
Military reservation	Suburban	Summer	0.30	[0.27, 0.37]	795	3	1.71	0.37	−0.30	0.50
Urban and center city										
Industrial	Urban and center city	Winter	0.35	[0.31, 0.40]	1788	8	2.48	0.36	0.36	−0.54
Residential	Urban and center city	Winter	0.35	[0.31, 0.38]	10402	42	2.21	0.37	0.39	−0.44
Commercial	Urban and center city	Winter	0.33	[0.29, 0.37]	11229	46	1.67	0.39	0.48	−0.43
Industrial	Urban and center city	Summer	0.28	[0.22, 0.34]	1881	8	0.92	0.40	0.19	−0.42
Industrial	Urban and center city	Spring	0.25	[0.20, 0.29]	1875	8	0.01	0.44	0.05	−0.42

confirms that, once physically inconsistent days are removed, the ranking is driven by the location of the fine–coarse partition rather than by tail artifacts.

For context-specific decisions, Table 5 lists, within each location setting, the non-thin cells with positive IFS (up to five). In rural areas, the residential winter cell leads, followed by agricultural cells in spring, summer, and autumn, consistent with secondary fine aerosol from agricultural ammonia and with wintertime residential combustion (Burns et al., 2023; Jandacka et al., 2024). In suburban areas, all built environments are fine-dominated in winter, and industrial and military reservation sites remain so in summer. In urban and center-city settings, the industrial, residential, and commercial winter cells lead, and only industrial sites retain positive values outside winter, suggesting a combustion-related fine component that persists across seasons near industrial sources (Cummings et al., 2021).

Table 6
Land-use contrasts of the FDI within each location setting and season, computed on station–season–year means. Groups list the land uses with at least three stations (number of stations in parentheses; Com. = commercial, Res. = residential, Ind. = industrial, Agr. = agricultural, For. = forest, Mil. = military reservation). H and ε^2 refer to the Kruskal–Wallis test, with Holm adjustment across the 48 omnibus tests; the last three columns give the pairwise contrast with the largest Cliff's δ and its Holm-adjusted Mann–Whitney p-value.

Location setting	Season	Groups	Units	H	ε^2	p_{Holm}	Largest contrast	δ	p_{Holm}
Rural	Winter	Com. (6), Res. (9), Ind. (4), Agr. (16)	89	12.43	0.141	0.235	Agricultural > Industrial	0.57	0.035
Rural	Spring	Com. (6), Res. (9), Ind. (4), Agr. (17)	103	9.45	0.093	0.667	Agricultural > Industrial	0.40	0.168
Rural	Summer	Com. (6), Res. (9), Ind. (4), Agr. (17), For. (3)	117	19.64	0.169	0.024	Forest > Industrial	0.94	0.003
Rural	Autumn	Com. (6), Res. (9), Ind. (4), Agr. (17), For. (3)	111	13.71	0.125	0.298	Forest > Industrial	0.80	0.024
Suburban	Winter	Com. (22), Res. (70), Ind. (7), Mil. (3)	281	5.05	0.018	1.000	Industrial > Military res.	0.32	0.865
Suburban	Spring	Com. (22), Res. (70), Ind. (7), Mil. (3)	295	5.77	0.020	1.000	Industrial > Residential	0.29	0.211
Suburban	Summer	Com. (22), Res. (70), Ind. (7), Mil. (3)	301	18.69	0.062	0.014	Industrial > Residential	0.49	0.001
Suburban	Autumn	Com. (22), Res. (70), Ind. (7), Mil. (3)	302	7.26	0.024	1.000	Industrial > Residential	0.31	0.116
Urban and center city	Winter	Com. (46), Res. (42), Ind. (8)	259	0.97	0.004	1.000	Residential > Commercial	0.07	1.000
Urban and center city	Spring	Com. (46), Res. (42), Ind. (8)	277	2.54	0.009	1.000	Industrial > Commercial	0.15	0.524
Urban and center city	Summer	Com. (46), Res. (42), Ind. (8)	280	3.31	0.012	1.000	Industrial > Commercial	0.24	0.201
Urban and center city	Autumn	Com. (46), Res. (42), Ind. (8)	286	1.66	0.006	1.000	Residential > Commercial	0.09	0.600

Rankings alone do not quantify separations between land uses, and Table 6 reports the formal contrasts on station–season–year units. After Holm correction across the 48 omnibus tests, land-use differences in the FDI are significant only in rural summer ($\varepsilon^2 = 0.17$, $p_{\mathrm{Holm}} = 0.024$) and suburban summer ($\varepsilon^2 = 0.06$, $p_{\mathrm{Holm}} = 0.014$). The largest effects involve forest sites, whose FDI exceeds that of industrial, commercial, and residential rural sites in summer ($\delta = 0.88$–0.94, large) and autumn ($\delta = 0.77$–0.80), and suburban industrial sites, which exceed residential ($\delta = 0.49$, large) and commercial ($\delta = 0.43$, medium) sites in summer. In urban and center-city settings, the effect sizes are negligible to small ($\varepsilon^2 \leq 0.012$; $|\delta| \leq 0.24$), so that commercial, residential, and industrial urban monitors share essentially the same fine–coarse partition in every season. This conclusion differs from what daily-level tests would suggest, because with tens of thousands of serially correlated days almost any difference becomes "significant"; it agrees with the companion station–season analysis, in which land-use contrasts were generally modest relative to seasonality (Sierra-Porta et al., 2026). For $PM_{2.5}$ concentrations, in contrast, land-use differences reach significance in urban summer ($\varepsilon^2 = 0.09$, $p_{\mathrm{Holm}} < 0.001$) and spring ($\varepsilon^2 = 0.05$, $p_{\mathrm{Holm}} = 0.024$), which indicates that land use modulates the level of fine particles more than their share of PM_{10}.

3.4. Robustness of the fine–coarse fingerprints

Table 7 summarizes the sensitivity of the IFS ranking. Across the 625 weight combinations, the median Kendall's τ with the nominal ranking is 0.88 (minimum 0.70), and the median overlap of the top-10 list is complete (minimum 0.8), whereas the bottom of the ranking, where cells are less separated, is somewhat more sensitive (median overlap 0.8, minimum 0.7). The first principal component of the standardized features explains 71% of their variance and correlates strongly with the IFS (Spearman $\rho = 0.88$); its loadings are dominated by the mean FDI (+0.66) and the mean r_0 (−0.59), which shows that the index is aligned with the main fine–coarse axis of the data. The component also loads positively on the skewness of r_0 (+0.46), because the bounded r_0 has more room for right-tail excursions where it is low, whereas the IFS deliberately penalizes those excursions as coarse bursts; consistently, the variant with positive shape weights yields the lowest agreement ($\tau = 0.72$). A stricter 90% completeness threshold (199 stations) and the retention of physically inconsistent days leave the ranking largely unchanged ($\tau = 0.90$ and 0.89; Spearman $\rho = 0.98$ in both cases).

The composition of the league tables is more sensitive to the treatment of thin cells than to the weighting. A tighter interval tolerance removes several winter hotspots with heterogeneous stations, whereas dropping the interval criterion lets sparse forest and agricultural winter cells enter the top of the ranking; in both cases the values of the IFS

Table 7
Robustness of the IFS ranking. Kendall's τ compares the nominal and alternative IFS values over the cells that are not thin under either specification; the overlaps compare each specification's own top-10 and bottom-10 lists with the nominal lists.

Specification	Kendall's τ	Top-10 overlap	Bottom-10 overlap
Weight grid (625 combinations), median [minimum]	0.88 [0.70]	1.0 [0.8]	0.8 [0.7]
Equal weights $(1, -1, -1, -1)$	0.80	0.9	0.8
Location terms only $(1.5, -1, 0, 0)$	0.84	1.0	0.8
Shape terms with positive sign $(1.5, -1, +0.5, +0.5)$	0.72	0.9	0.6
Mean FDI only	0.86	0.9	0.8
First principal component (71% of variance)	0.71	0.9	0.6
Completeness ≥90% (199 stations)	0.90	0.9	0.9
Days with $PM_{2.5} > PM_{10}$ retained (r_{12} capped at 1)	0.89	1.0	0.9
Thin-cell interval tolerance 0.10 (25 non-thin cells)	1.00	0.6	0.7
No interval criterion for thin cells (48 non-thin cells)	1.00	0.4	0.8
Season–year 2023 only	0.75	0.6	0.7
Season–year 2024 only	0.79	0.8	0.6
Season–year 2025 only	0.81	0.7	0.7

are unchanged ($\tau = 1$), so that only the eligibility of cells differs. Recomputing the IFS separately for each season–year reproduces the nominal ranking with Spearman correlations of 0.91–0.93 ($\tau = 0.75$–0.81). Winter cells of built environments occupy 7 and 6 of the top-10 positions in 2024 and 2025, respectively, but only 4 in 2023, when the winter season covers only January and February and the summer was affected by smoke from the record Canadian wildfires (Wang et al., 2024); in that year, summer and spring cells of suburban and urban industrial sites rank higher. The qualitative hierarchy (wintertime fine dominance in built environments, coarse dominance in rural industrial and desert settings) is therefore robust, whereas the exact order of mid-ranked cells should not be over-interpreted.

3.5. Temporal complexity of daily PM series

Table 8 summarizes the complexity metrics by parameter. All size fractions occupy the high-entropy, low-complexity region of the complexity–entropy plane (Fig. 4): with $D = 4$ over the full record, the median permutation entropy ranges from 0.955 for $PM_{2.5}$ FRM/FEM to 0.964 for PM_c, and the statistical complexity from 0.055 to 0.044. Daily PM concentrations therefore behave as strongly stochastic, correlated-noise processes, as also reported for most environmental time series (Amato et al., 2020; Nikolopoulos et al., 2021; Ribeiro et al., 2017). They are not random, however: shuffled surrogates reach $H = 0.997$ and $C_{JS} = 0.004$, and the full-period entropy of 99.8–100% of the stations lies significantly below its surrogate distribution. Within individual seasonal windows, where only about 90 values are available, the ordinal structure is detectable in 48–59% of the windows, with the highest rates in summer for $PM_{2.5}$ FRM/FEM (70%) and the lowest in winter for all parameters (36–41%). Day-to-day winter variability is thus closer to random, consistent with the rapid alternation of stagnation and ventilation imposed by synoptic systems, whereas summer variability is more persistent.

The size fractions differ systematically in their ordinal structure. At collocated sites, $PM_{2.5}$ FRM/FEM is more ordered than PM_{10} ($r_{rb} = -0.35$ for H, 242 sites) and much more ordered than PM_c ($r_{rb} = -0.69$, 91 sites), and PM_{10} is more ordered than PM_c ($r_{rb} = -0.61$, 78 sites; all $p < 0.001$), whereas the two $PM_{2.5}$ methods are indistinguishable ($r_{rb} = -0.01$, 22 sites). The ordering $PM_{2.5} < PM_{10} < PM_c$ in permutation entropy is physically meaningful: fine particles are dominated by regionally mixed secondary aerosol and long-range transport, which vary smoothly with synoptic meteorology, whereas coarse particles respond to local, intermittent resuspension and wind events that decorrelate quickly. Within seasonal windows ($D = 3$), the same ordering appears with smaller effect sizes and remains significant after Holm correction only for the PM_{10}–PM_c contrast ($r_{rb} = -0.48$; Table 9), where the coarse fraction is also more skewed ($r_{rb} = -0.49$ for skewness). Figure 8a shows that the metrics capture partly distinct properties: with $D = 3$, H and C_{JS} are perfectly redundant ($\rho = -1.00$), so that the complexity–entropy plane collapses to a curve; SVD entropy and the Higuchi dimension are strongly related ($\rho = 0.71$), skewness and kurtosis almost interchangeable ($\rho = 0.96$), and sample entropy decreases with tail weight ($\rho = -0.59$ with skewness), whereas ordinal and amplitude-based irregularity are nearly independent ($\rho = 0.11$ between H and sample entropy). Reporting a subset of non-redundant metrics, as done below, avoids double counting the same information.

Table 8
Temporal-complexity metrics by parameter: medians [interquartile ranges] across season–year windows ($D = 3$) and across stations for the full-period ordinal analysis ($D = 4$). Ordered windows are those whose permutation entropy lies below the 5th percentile of 199 shuffled surrogates.

	$PM_{2.5}$ FRM/FEM	$PM_{2.5}$ non-FRM/FEM	PM_{10}	PM_c
Stations	751	171	420	91
Season–year windows	8664	1936	4782	1029
Ordered windows (%)	58.1	58.8	49.9	47.8
Permutation entropy H ($D = 3$)	0.965 [0.949, 0.978]	0.964 [0.948, 0.977]	0.969 [0.954, 0.981]	0.970 [0.955, 0.982]
Statistical complexity C_{JS} ($D = 3$)	0.033 [0.021, 0.047]	0.033 [0.022, 0.048]	0.029 [0.018, 0.043]	0.028 [0.018, 0.041]
Sample entropy	1.61 [1.33, 1.88]	1.50 [1.16, 1.79]	1.51 [1.21, 1.80]	1.58 [1.28, 1.82]
SVD entropy	0.943 [0.916, 0.964]	0.929 [0.901, 0.956]	0.951 [0.922, 0.975]	0.954 [0.929, 0.974]
Higuchi fractal dimension	1.84 [1.78, 1.89]	1.83 [1.76, 1.88]	1.85 [1.79, 1.90]	1.85 [1.80, 1.90]
Correlation dimension D_2	1.25 [1.15, 1.36]	1.28 [1.16, 1.39]	1.29 [1.18, 1.40]	1.26 [1.16, 1.38]
Skewness	1.17 [0.76, 1.85]	1.33 [0.86, 2.12]	1.31 [0.78, 2.32]	1.28 [0.80, 2.06]
Excess kurtosis	1.55 [0.30, 4.74]	2.10 [0.57, 5.94]	2.15 [0.38, 7.75]	1.93 [0.43, 6.44]
H ($D = 4$, full period)	0.955 [0.947, 0.962]	0.956 [0.946, 0.964]	0.961 [0.954, 0.970]	0.964 [0.957, 0.970]
C_{JS} ($D = 4$, full period)	0.055 [0.047, 0.064]	0.054 [0.044, 0.065]	0.047 [0.037, 0.055]	0.044 [0.037, 0.051]

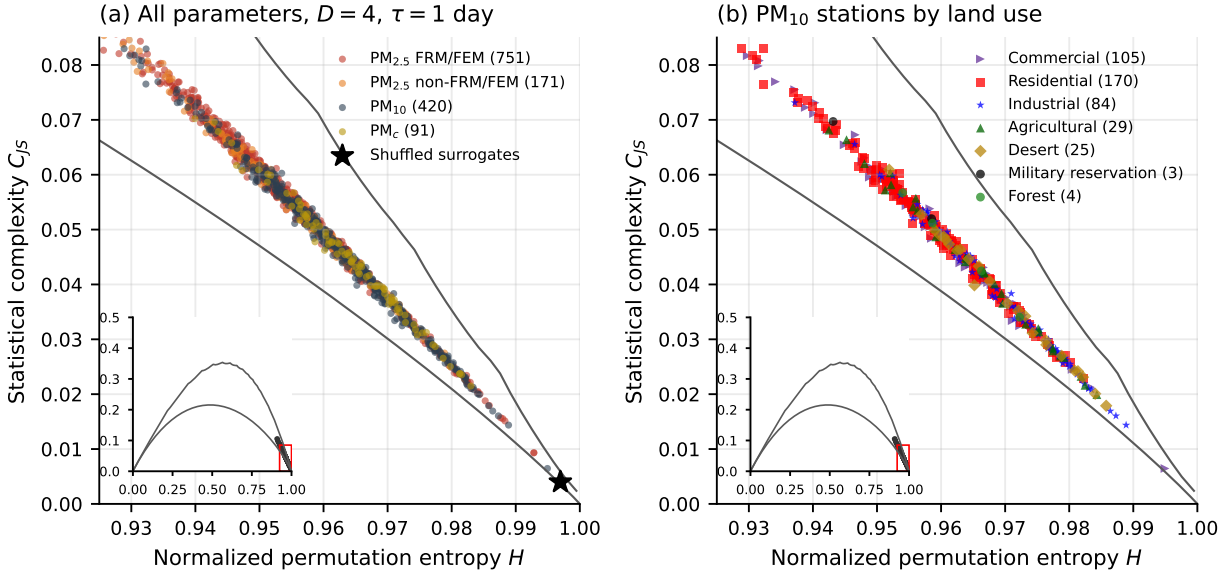


Figure 4: Complexity–entropy causality plane for the full 2023–2025 record of each station ($D = 4$, $\tau = 1$ day). Grey curves are the minimum and maximum statistical complexity attainable for a given entropy; the insets show the complete plane, with the zoomed region outlined in red. (a) All parameters, with the median of the shuffled surrogates (star). (b) PM_{10} stations colored by land use.

3.6. Seasonal and land-use structure of PM dynamics

Seasonality is the main source of variation in the dynamics of the fine fraction (Fig. 5 and Table 9). For $PM_{2.5}$ FRM/FEM, summer windows are smoother and more persistent than winter windows, with lower sample entropy (median 1.36 versus 1.71), lower SVD entropy (0.922 versus 0.956), and lower Higuchi dimension (1.79 versus 1.88; Kendall's $W = 0.19$–0.32), and they are also more heavy-tailed (excess kurtosis 3.36 versus 1.06). The combination of smooth background variability and rare, intense excursions is the signature of multi-day episodes, such as heat-driven secondary formation and wildfire smoke; the latter was exceptional during the 2023 Canadian fire season, which affected large parts of the United States (Wang et al., 2024; Butler and Hanigan, 2024). The non-FRM/FEM $PM_{2.5}$ series reproduce the same pattern, with an even stronger seasonal contrast in sample entropy ($W = 0.34$). PM_{10}

and PM_c show weaker seasonality ($W \leq 0.22$), with the smoothest trajectories in autumn and no significant seasonal change in sample entropy.

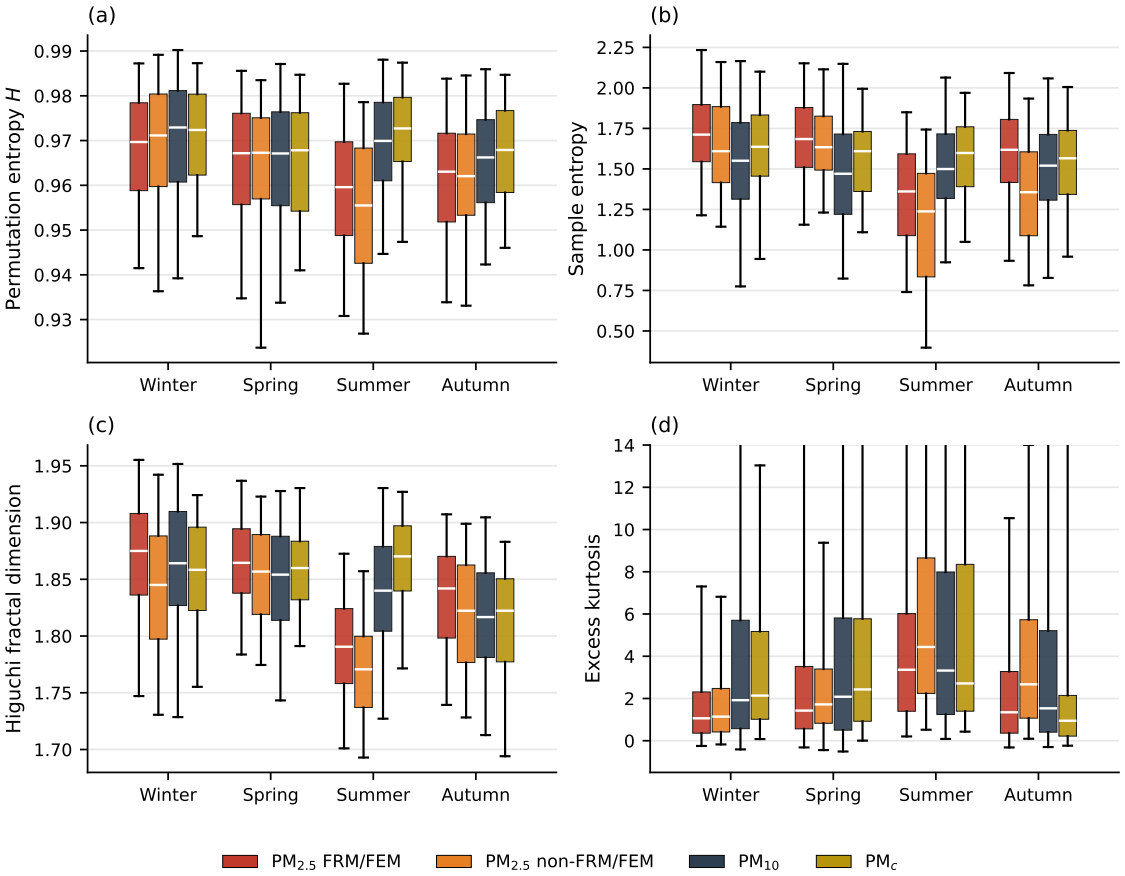


Figure 5: Seasonal distribution of station-median complexity metrics by parameter: (a) permutation entropy ($D = 3$), (b) sample entropy, (c) Higuchi fractal dimension, and (d) excess kurtosis. Boxes show interquartile ranges and whiskers the 5th–95th percentiles; in panel (d) the axis is truncated at 14.

Land use, in contrast, has almost no influence on $PM_{2.5}$ dynamics but a moderate influence on PM_{10} dynamics (Fig. 6). For $PM_{2.5}$ FRM/FEM, the land-use effect sizes are negligible to small ($\varepsilon^2 \leq 0.047$), and the largest of them involve the tail statistics, with heavier tails at agricultural and forest sites (median kurtosis 2.2 and 2.6 versus 1.0–1.5 elsewhere), consistent with episodic smoke and secondary aerosol events in rural regions. For PM_{10}, land use explains 13–15% of the rank variance of the ordinal, amplitude, and tail metrics ($\varepsilon^2 = 0.125$–0.147; 0.043 for D_2). Desert, industrial, and agricultural monitors, most of them rural, combine higher permutation entropy (0.971–0.976 versus 0.966–0.967 at commercial and residential sites) with lower sample entropy (1.20 at desert sites versus 1.55) and much heavier tails (median excess kurtosis 7.8 at desert sites versus 1.5–1.6). The apparent contradiction between ordinal and amplitude-based irregularity is informative: in a series dominated by rare, large dust or resuspension events, the standard deviation is inflated by the bursts, so that the background days fall within the tolerance of sample entropy and look regular, while the order of consecutive days remains almost random. The same contrast appears along the urbanicity gradient, where rural PM_{10} is more random in its ordinal structure and more heavy-tailed than suburban and urban PM_{10} ($\varepsilon^2 = 0.12$–0.14). For PM_c, the land-use contrasts are not significant, although the small number of non-urban PM_c monitors limits this comparison.

Table 9
Effect sizes for the temporal-complexity metrics: Kendall's W for the seasonal contrast within stations (Friedman test), ε^2 for the land-use contrast among station medians (Kruskal–Wallis test), and the matched-pairs rank-biserial correlation r_{rb} for collocated size fractions (Wilcoxon signed-rank test; negative values indicate lower values in the first fraction). Window-level metrics with $D = 3$. Holm-adjusted significance within each family: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

	Seasonality (W)			Land use (ε^2)		Paired fractions (r_{rb})	
Metric	$PM_{2.5}$ FRM/FEM	PM_{10}	PM_c	$PM_{2.5}$ FRM/FEM	PM_{10}	$PM_{2.5}$ vs PM_{10}	PM_{10} vs PM_c
H	0.064***	0.036***	0.060**	0.008	0.127***	−0.17	−0.48**
Sample entropy	0.194***	0.005	0.018	0.036**	0.125***	+0.12	+0.22
SVD entropy	0.319***	0.118***	0.169***	0.014	0.143***	−0.38***	−0.37
Higuchi dimension	0.318***	0.096***	0.215***	0.010	0.129***	−0.14	−0.28
D_2	0.136***	0.069***	0.097***	0.034**	0.043	−0.32***	+0.14
Excess kurtosis	0.103***	0.039***	0.089***	0.047***	0.128***	−0.09	−0.39

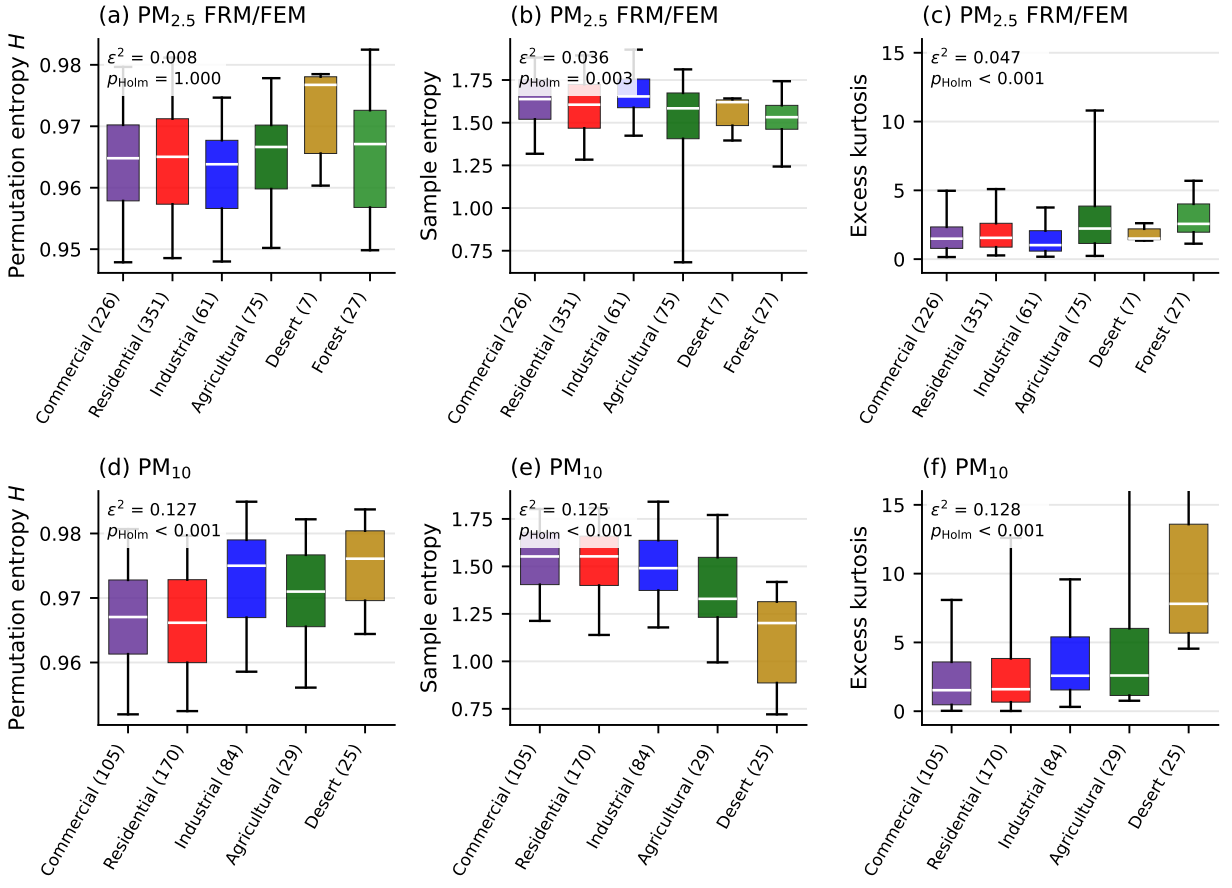


Figure 6: Station-median complexity metrics by land use for $PM_{2.5}$ FRM/FEM (a–c) and PM_{10} (d–f): permutation entropy ($D = 3$), sample entropy, and excess kurtosis. Only land uses with at least five stations are shown (number of stations in parentheses). Boxes show interquartile ranges and whiskers the 5th–95th percentiles. The Kruskal–Wallis effect size ε^2 and the Holm-adjusted p-value are given in each panel.

3.7. Spatial organization of PM dynamics

Maps of the station-median permutation entropy show coherent regional patterns (Fig. 7a,b). For $PM_{2.5}$, the most ordered dynamics occur in California and in the southeastern and eastern United States, whereas the most random

Table 10
Spearman correlations between the station mean FDI and the station-median complexity metrics of the collocated $PM_{2.5}$ FRM/FEM and PM_{10} series (240 stations), and partial correlations controlling for latitude and longitude. †Holm-adjusted $p < 0.05$.

	$PM_{2.5}$ FRM/FEM		PM_{10}	
Metric	ρ_s	**Partial** ρ_s	ρ_s	**Partial** ρ_s
H ($D = 4$, full period)	$-0.31^{\dagger}$	−0.34	−0.14	−0.27
C_{JS} ($D = 4$, full period)	$+0.33^{\dagger}$	+0.34	+0.17	+0.29
Sample entropy	$+0.21^{\dagger}$	+0.19	$+0.33^{\dagger}$	+0.24
SVD entropy	−0.05	−0.37	$-0.24^{\dagger}$	−0.45
Higuchi fractal dimension	+0.10	−0.29	−0.11	−0.40
Skewness	−0.04	−0.21	$-0.32^{\dagger}$	−0.39
Excess kurtosis	−0.11	−0.26	$-0.35^{\dagger}$	−0.37

dynamics occur in the northern Plains and the Mountain West; for PM_{10}, the highest entropies concentrate in the arid Southwest and in Wyoming, where many rural industrial and desert monitors are located. Ward clustering of the $PM_{2.5}$ dynamical profiles summarizes this organization (Fig. 7c and Fig. 8b). Cluster 1 (120 stations; median longitude −118.5°) is concentrated along the Pacific coast, mainly in California, and has the most structured dynamics ($H = 0.958$, $C_{JS} = 0.039$) and the smoothest trajectories (SVD entropy 0.913, Higuchi dimension 1.79); cluster 2 (134 stations; −98.5°) groups the northern Plains, the Mountain West, the upper Midwest, and southern Florida and has the most random ordinal dynamics ($H = 0.969$, $C_{JS} = 0.029$) together with the lowest sample entropy (1.44); and cluster 3 (497 stations; −86.1°) covers most of the East and the Gulf coast and has the highest sample entropy (1.67). The partition is only moderately separated (silhouette width 0.22 for $k = 3$; median adjusted Rand index of 0.40 between the full solution and solutions from 80% subsamples), so the profiles are better described as a continuum with a dominant west–east gradient than as discrete types. In every partition we examined, however, the groups are unrelated to land use (adjusted Rand index −0.02; Cramér's $V = 0.13$) and to location setting (adjusted Rand index 0.00; $V = 0.08$), whereas they differ strongly in longitude (Kruskal–Wallis $H = 173$, $p < 0.001$). $PM_{2.5}$ dynamics are therefore organized by regional meteorology and source mixtures rather than by the local land-use label of the monitor. The persistence of station rankings between years is moderate (Spearman $\rho = 0.17$–0.41 for $PM_{2.5}$ and 0.23–0.53 for PM_{10}, highest for the Higuchi dimension of PM_{10}), which indicates that interannual variability, including the 2023 smoke episodes, is an important component of the dynamics at individual sites.

3.8. Linking fine dominance and temporal dynamics

The two layers of the analysis are related at the station level (Table 10 and Fig. 9). Stations with higher mean FDI have more structured $PM_{2.5}$ dynamics, with lower full-period permutation entropy ($\rho_s = -0.31$) and higher statistical complexity ($\rho_s = 0.33$), and less bursty PM_{10}, with lower skewness ($\rho_s = -0.32$) and kurtosis ($\rho_s = -0.35$) and higher sample entropy ($\rho_s = 0.33$). These associations persist after controlling for latitude and longitude (|partial ρ_s| = 0.24–0.39), so they are not a by-product of the regional gradients described above. Fine-dominated regimes therefore combine a larger share of fine particles with smoother, more persistent fine-particle dynamics and a PM_{10} signal without large coarse bursts, whereas coarse-dominated regimes combine a lower FDI with intermittent, heavy-tailed PM_{10} variability. The size partition and the temporal dynamics are thus two complementary expressions of the same source regime, and their joint use separates sites that a single ratio would treat as equivalent.

3.9. Interpretability, implications, and limitations

The Fine Dominance Index is interpretable by design. Because it is a bounded, convex transformation of $PM_{2.5}/PM_{10}$, larger values correspond to stronger fine-particle dominance, typically associated with combustion and secondary aerosol, while smaller values indicate coarse dust or resuspension. The composite IFS standardizes cell-level summaries into an integrated fingerprint for ranking, without replacing the physically meaningful FDI, and the complexity layer adds information that neither index contains: whether the concentrations of a given fraction evolve smoothly or intermittently. In practice, we recommend reporting the ranked IFS for prioritization, the FDI mean with its station-block interval for interpretation, and, for monitors with suspected coarse influence, the permutation entropy and kurtosis of PM_{10} as indicators of episodic coarse sources.

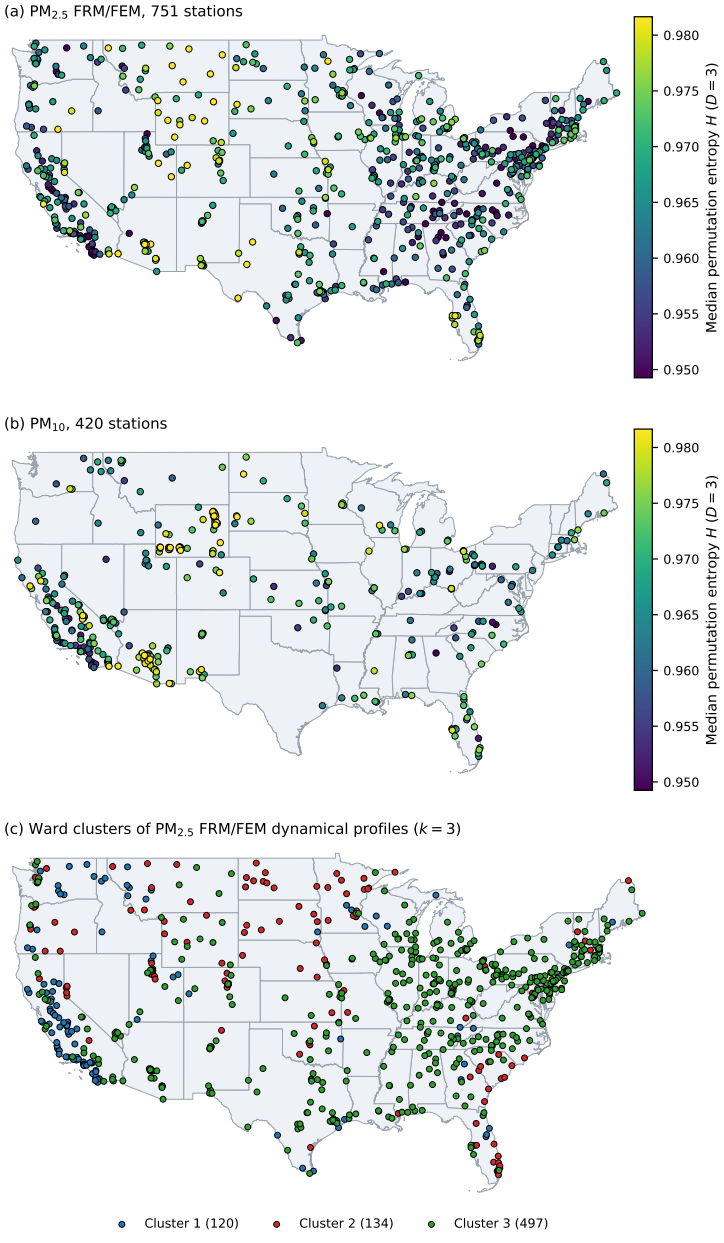


Figure 7: Spatial organization of PM dynamics. (a) Station-median permutation entropy ($D = 3$) of $PM_{2.5}$ FRM/FEM and (b) of PM_{10}, on a common color scale (5th–95th percentiles). (c) Ward clusters ($k = 3$) of the standardized dynamical profiles (permutation entropy, sample entropy, SVD entropy, and Higuchi dimension) of $PM_{2.5}$ FRM/FEM stations, numbered from west to east.

Several limitations should be noted. Land-use labels and location settings are categorical proxies that cannot capture within-class heterogeneity, proximity to specific sources, or micro-siting effects, and the regulatory network is not a random sample of U.S. environments: desert, forest, and military reservation monitors are few in the fine–coarse network, so the corresponding cells are flagged as thin, and the coarse-dominated regime is best documented by the PM_{10} complexity network. The indices rely on mass concentrations only and do not resolve chemical composition, hygroscopic growth, or aerosol acidity. $PM_{2.5}$ and PM_{10} are measured by different instruments, so that their ratio inherits instrumental uncertainty, and the 1 µg m^{-3} resolution of PM_{10} creates ties that we resolved at random. Seasonal

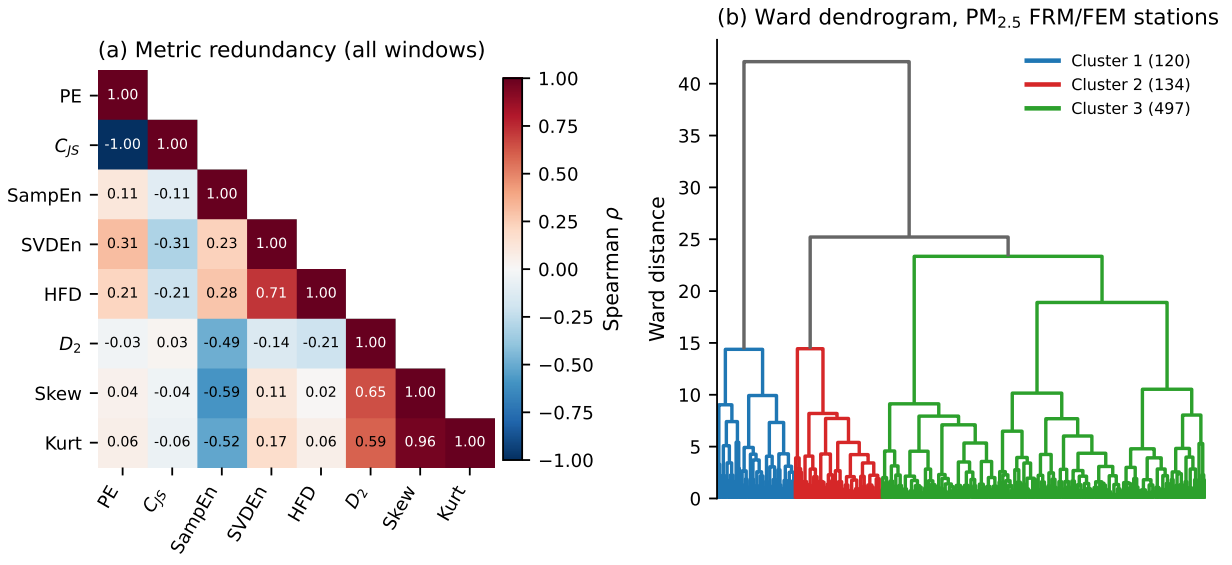


Figure 8: (a) Spearman correlations among the complexity metrics across all season–year windows (PE: permutation entropy; C_{JS}: statistical complexity; SampEn: sample entropy; SVDEn: SVD entropy; HFD: Higuchi fractal dimension; D_2: correlation dimension; Skew: skewness; Kurt: excess kurtosis). (b) Ward dendrogram of the $PM_{2.5}$ FRM/FEM stations, with the three clusters mapped in Fig. 7c.

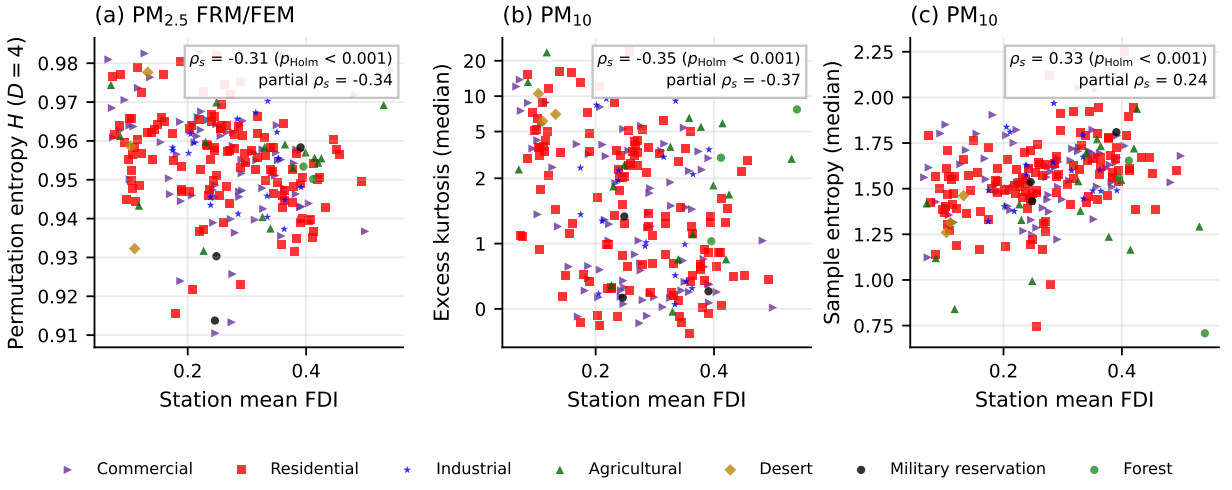


Figure 9: Station mean FDI versus (a) the full-period permutation entropy of $PM_{2.5}$ FRM/FEM ($D = 4$), (b) the median excess kurtosis of PM_{10} (symmetric logarithmic axis), and (c) the median sample entropy of PM_{10}, for the 240 stations of the fine–coarse network, colored by land use. Each panel reports the Spearman correlation with its Holm-adjusted p-value and the partial correlation controlling for latitude and longitude.

windows of about 90 days limit the embedding dimension of ordinal analyses ($D = 3$ within seasons) and constrain the correlation dimension, which is bounded by the embedding dimension of 2 and should be read as a low-dimensional geometric descriptor rather than as an estimate of an attractor dimension. Finally, seasons are defined climatologically and meteorological covariates were not modeled, so that the interpretation of seasonal and regional contrasts in terms of boundary-layer dynamics, smoke, and dust remains inferential.

Despite these caveats, the convergence of distributional evidence, cell-wise fingerprints, station-level tests, and temporal-complexity metrics supports a consistent message: the fine–coarse partition of U.S. particulate matter is

controlled first by season and region and only secondarily by land use, and coarse-dominated regimes are characterized as much by the intermittency of their dynamics as by their mass partition. A compact set of complementary metrics, namely the FDI at the sample level, cell-wise summaries with uncertainty, a standardized composite index, and a few non-redundant complexity descriptors, yields a reproducible, source-informed fingerprint that is stable to reasonable modeling choices and directly usable for season- and context-specific air-quality management.

4. Conclusions

Relying on a single $PM_{2.5}/PM_{10}$ ratio or on fixed thresholds is insufficient to separate particulate regimes across heterogeneous U.S. contexts. By combining a bounded, sample-level Fine Dominance Index, cell-wise seasonal fingerprints with station-block uncertainty, station-level nonparametric contrasts, and a temporal-complexity layer, we obtain a compact and interpretable characterization of fine versus coarse dominance for 2023–2025. Fine dominance is primarily seasonal: it peaks in winter at most monitors, and the robust hotspots are winter cells in suburban and urban built environments, whereas coarse dominance prevails at rural industrial sites outside winter and at the few desert monitors. Once pseudo-replication is avoided, land-use differences within a setting are mostly small, with the exception of fine-dominated forest sites and suburban industrial sites.

Daily PM concentrations behave as high-entropy, correlated-noise processes that are nevertheless significantly more ordered than shuffled surrogates. Ordinal entropy increases from $PM_{2.5}$ to PM_{10} to PM_c, summer fine-particle dynamics are smoother but more heavy-tailed, and land use shapes the dynamics of PM_{10}, with intermittent, heavy-tailed variability at desert, industrial, and agricultural sites, but not those of $PM_{2.5}$, which vary mainly with geography. Fine-dominated stations display more structured $PM_{2.5}$ and less bursty PM_{10} dynamics, which shows that mass partition and temporal dynamics are complementary signatures of the same source regimes. Sensitivity analyses on index weights, completeness thresholds, quality-control rules, thin-cell criteria, and individual years indicate that the qualitative hierarchy is stable, although the exact membership of the ranked tables depends on how sparse cells are treated and on the year considered.

Practically, the findings support season- and context-specific air-quality management: wintertime mitigation of combustion and secondary fine aerosol in built environments, and abatement of intermittent coarse sources, such as unpaved roads, construction, mining, and exposed soils, where PM_{10} dynamics are bursty. Future work should integrate chemical speciation and receptor models to validate the fingerprints; incorporate meteorological covariates, boundary-layer data, and smoke and dust event flags to attribute seasonal and regional contrasts; expand coverage for sparse classes such as desert, forest, and military areas; analyze hourly data, which would allow higher embedding dimensions and multiscale complexity measures; extend the temporal scope to test the persistence of the fingerprints; and package the workflow into an operational tool that updates automatically from EPA AirData for routine screening and policy support.

Data and code availability

All raw data are openly available from the U.S. EPA AirData pre-generated daily summary archives for 2023–2025 (`https://aqs.epa.gov/aqsweb/airdata/download_files.html`). The analysis pipeline, which downloads the data, applies the screening and quality-control rules, and reproduces every table and figure of this article, is publicly available at `https://github.com/IsabellaArrieta/atmospheric-pollutants-analysis`.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with code review, the reanalysis scripts, and language editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

D. Sierra-Porta: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Supervision. **Isabella S. Arrieta-Guardo:** Methodology, Software, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Visualization. **Jose F. Gonzalez-Ortiz:** Methodology, Software, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Visualization. **Eduardo A. Negrin-Perez:** Methodology, Software, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Visualization.

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